

## NanoFace-2N(P)

### Facial Recognition System

Compact terminal with ultra-fast subject finding and local authentication



#### APPLICATION

*Compact, wall-mountable terminal for high throughput access control, time & attendance, and ID management applications*

### Product Description

NanoFace-2N(P)™ is an advanced face recognition terminal that provides highly accurate face recognition in a compact and cost-effective embedded system. Employing the latest high-performance ARM processor, the NanoFace-2N features an embedded face recognition module with its own imaging and algorithm co-processor for ultra-fast capture and processing of face images.

This architecture allows for the utilization of the latest and most powerful, but complex, face algorithms available. Now, the NanoFace-2N(P) face recognition system not only provides a fully touchless and contactless user experience, but embedded solution matching accuracy is unparalleled! In authentication mode, capture speeds are virtually “walk-through” while very fast image and template processing times make authentication decisions seem instantaneous.

The NanoFace-2N(P) system is compact, with dimensions of only 100 x 200 x 36 mm, and can readily be mounted on any wall, desktop, kiosk

front-plate, or turnstile.

The user interface is effortless and intuitive. The subject's face can be found at up to 2 meters from the NanoFace-2N(P) system, with an active capture range from 0.4 to 2 meters. The NanoFace-2N(P) incorporates a highly innovative digital zoom feature that enlarges the subject's face image during capture, and automatically tilts the display centering as the subject moves within the capture volume.

Eyelock's proprietary deep-learning based face-detector also enhances operation in a wide range of ambient lighting for deployment flexibility. User positioning instructions are clearly shown on the display to guide subject movements, while authentication and authorization decisions are presented immediately in clear notations around the subject's face.

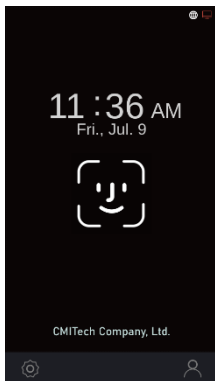
The system also provides color-based cues for face finding, fake face detection, and mask detection, in addition to positive identification (or authentication), and identification rejection.

The on-board white and NIR illuminators expand the deployment flexibility of NanoFace-2N(P), making it the ideal choice for advanced face recognition solutions across a diverse range of access control and time & attendance installations.

## Simple and Intuitive Subject Positioning

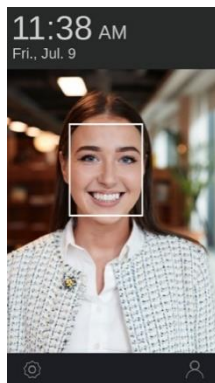
NanoFace-2N(P) detects and displays the subject's face over 2 meters from the system on the high-resolution color display. The subject will simply and naturally walk toward the face capture range of 0.4 to 2 meters. Once the system recognizes the face, the result will be displayed immediately with a colored indication box over the subject's face image.

## Access Control Authentication (Verification)



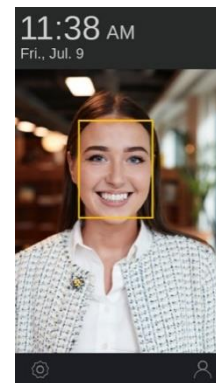
### LAUNCHER VIEW

Home Page for start-up and administrator control.



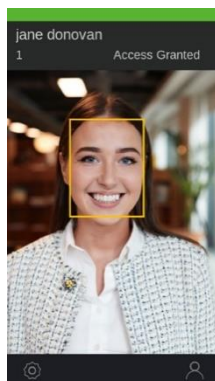
### FACE DETECTION STARTS

White box around face shows which subject has been detected and is being actively processed



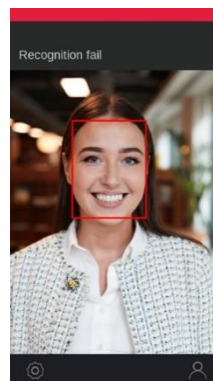
### FACE RECOGNITION STARTS

Display color box around face indicates normal face (yellow) fake face (orange), or masked face (green)



### IDENTIFICATION SUCCESS!

Subject is recognized and authorized: name and ID are shown.



### IDENTIFICATION REJECTION

No match against internal, on-board data base.

## Key Features

### Feature

State of art optical design

Dual face cameras

Wide angle face imaging

Advanced real time subject tracking with simple user instructions

Intuitive user interface

Face matching engine

Display of authentication (matching) results

Language support

Large on-board (embedded) face template data base

High speed face matching

Widest range of lighting conditions

Standard multi-band RFID reader

HID™ compatible card reader

Live face detection

Mask detection

### User Advantages

Optical design utilizes highest quality optics and a very fast shutter speed, allowing the system to exceed industry norms for image quality.

Two cameras—one in visible and one in NIR—optimize face detection in the lowest of ambient lighting conditions and provide for superior live face detection.

Outstanding capture volume allows face recognition at distance range (stand-off) of 0.4 to 2.0 meters.

NanoFace-2N(P) accurately locates the subject face in real time.

Modeled after latest smart-phone user displays, the user experience will be intuitive and effortless for virtually all subjects.

Latest generation neural network-based encoding and matching algorithm delivers highest levels of matching accuracy. Includes integrated dual cameras with algorithm co-processor for fastest processing speeds.

Automatic display of matching results, positioned adjacent to subject's face.

English, Korean, Simplified Chinese, Traditional Chinese, Japanese, Arabic, Spanish, Italian, Turkish, French

Stores up to 20,000 active face templates on-board in 1:N recognition (identification) mode.

Up to 50,000 matches per second on-board.

Embedded illuminators in both white and NIR ranges expand use in adverse ambient lighting environments.

MiFare, DesFire EV1/EV2/EV3, FeliCa card support with standard embedded ISO/ IEC 14443 reader.

HID Ominkey™ embedded card reader optional, for compatibility with range of HID iClass™ cards

YES, included in proprietary algorithm

Yes

## Feature

RFID

PoE (power over Ethernet)

Powerful and simple SDK

Full range of deployment options

## User Advantages

13.56 MHZ operating frequency

Applies to NFACE2NP Only

Based on Eyelock's proven high level SDK architecture and code, the NanoFace-2N(P) is simple to integrate. All APIs reside on host-side application in Windows C++, Windows C# (.NET) and Linux OS, so no device level programming is necessary. Reference code supplied with SDK.

Standard connections include Ethernet, dual Wiegand in and out, GPI, RS-485, dry contact relay, tamper detection, factory reset button, SIM socket

## Technical Specifications

|                                                   |                                                                                                                                      |
|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| CPU                                               | Quad core ARM with NPU                                                                                                               |
| Memory                                            | 2 GB RAM<br>8 GB flash                                                                                                               |
| Number of cameras                                 | Two (2)                                                                                                                              |
| Illumination                                      | One visible (white) and one near-infrared (NIR) LED for optimal face detection and operation in low ambient light conditions         |
| Dimensions                                        | 100 x 200 x 36 mm                                                                                                                    |
| Weight                                            | 420 g (without standard wall mount plate)                                                                                            |
| Display                                           | 5.0 inch (nominal), touch                                                                                                            |
| Operating capture range                           | 40 cm to 200 cm<br>Configurable maximum operating distance                                                                           |
| User height range                                 | 140 cm to 210 cm (with system installed at 120 cm)                                                                                   |
| Face extraction (encoding) and matching algorithm | YES, on-board algorithm functions included as standard                                                                               |
| Authentication speed                              | Within 1.0 second                                                                                                                    |
| Enrollment speed                                  | Within 5.0 second                                                                                                                    |
| Fake face detection                               | YES                                                                                                                                  |
| Data base size, on-board                          | Maximum 20,000 subjects in 1:N mode (authentication)                                                                                 |
| Text Log                                          | 5,000,000                                                                                                                            |
| Image Log                                         | 10,000                                                                                                                               |
| Audio In/Out                                      | Embedded microphone and built-in speaker                                                                                             |
| Power requirement                                 | 15 to 24 V DC, 2A. Maximum power consumption 30 watts.<br>Universal power supply included.                                           |
| RF card reader                                    | Standard ISO/IEC 14443 A/B, MiFare, DesFire EV1/EV2/EV3, FeliCa.                                                                     |
| HID iClass                                        | Optional OMNIKEY chipset                                                                                                             |
| RFID                                              | 13,56 MHZ operating frequency                                                                                                        |
| I/O connections                                   | RJ45 for Ethernet, Wiegand In and Out, GPI (3), dry contact relay, RS485, tamper, factory reset, SIM socket (optional), USB 2.0 host |
| OSDP compatibility                                | YES<br>1 channel master or slave through RS-485 port                                                                                 |
| PoE - NFACE-2NP Only                              | Optional (IEEE 802.3af)                                                                                                              |
| Operating temperature                             | -20° C to 50° C                                                                                                                      |
| Operating humidity                                | 10% to 95% non-condensing                                                                                                            |
| Certifications                                    | ISO 9001, CE, FCC, KC, UKCA                                                                                                          |